sea; strict control of industrial pollution source discharge, completion of illegal and unreasonable clean-up of sewage outlets; promotion of agriculture, rural and urban pollution prevention and control; and reduction of land-based pollutants into the sea. The marine pollution control actions include: implementing marine aquaculture pollution control; implementing ship and port pollution control; carrying out comprehensive improvement of the fishing port environment; marine garbage pollution prevention; and establishing a division of responsibilities and coordination mechanism for land and sea planning. The ecological protection and restoration actions include ecological protection of coastal zones, delineation and strict observance of the red line of marine ecological protection in the Bohai Sea, comprehensive regulation and restoration of estuaries and bays, comprehensive management and restoration of coastal and shoreline, and conservation of marine living resources. Environmental risk prevention actions include the implementation of land-based emergency environmental incident risk prevention; implementation of marine oil spill risk prevention; marine ecological disaster warning and emergency response.

(h) Blue Economy Application Cases (Management Policy)

Drawing from science-based products and services that can underpin the development of the Blue Economy, and China's marine integrated management based on ecological environment, the paper classifies all the cases into disaster prevention, pollution prevention, Marine industry support, system platform, and ecological restoration.

(i) Disaster Prevention

(i) Assessing Harmful Algal Bloom Evolution in EU Atlantic Shelf Seas

Aquaculture is identified as a Blue Growth priority to ensure the sustainable supply of seafood to help meet the increased food demands of the growing global population. Harmful Algal Blooms (HABs) are a recognized global problem with annual losses to aquaculture running into billions of Euros (Bernard, et al., 2014). The AtlantOS HAB use case focuses on creating a weekly HAB bulletin for three European study areas in Norway, Ireland, and Spain. Experts who prepare the bulletins use the *in situ* ocean observing system, satellite data, and available numerical marine

hydrodynamic modeling to provide a science-based product to indicate the current HAB status in areas of interest, accompanied by text describing the likely HAB occurrences in the days ahead (Cusack et al., 2018); example bulletins). Efforts in other parts of the world, the importance of co-development, and future expectations are discussed in Anderson et al (2019). In Europe, a key driver for HAB bulletins is to support industry on-site management decisions, such as optimizing farm practices and planning business activities to mitigate/reduce shellfish and finfish mortalities due to HABs.

(ii) Assessing and Mapping Ocean Hazards Related to Coastal Flooding and Storm Surges

Coastal flooding is one of the major challenges of global climate change for humanity. It is estimated that by 2070, approximately 150 million people and $35,000 billion of assets will be exposed to a 1 in 100 year flood event (WMO, 2021). Storm surges and oceanic waves are the major cause of extreme sea levels and devastating coastal impacts along many coastlines around the world with significant human and economic costs. To improve our ability to assess the potential change of storm surge, it is critical to have a well-established baseline of the storm surge climate, based on consistent techniques. AtlantOS, an oil spill hazard maritime transportation estimator in the Atlantic Ocean basin coast, is laying the framework for a complementary international effort; developing a global storm surge climate has been adopted as a project under the Intergovernmental Oceanographic Commission (IOC)/World Meteorological Organization (WMO), Joint WMO-IOC Commission for Oceanography and Marine Meteorology (JCOMM) Expert Team for Waves and Coastal Hazards. Statistical methods for both tide gauge data and multi-decadal runs of hydrodynamical numerical models are the "main" activity for the Sendai Framework of Disaster Risk Reduction. In this use case improvements were made to estimate extreme sea levels around the Atlantic and more widely with a new understanding gained about how storm surges and high tides interact, providing proof that any storm surge can occur on any tide (Williams et al, 2016). The science-based products developed in this use case will help to enhance the safety of coastal communities by supporting the decision-making process related to planning coastal defenses and emergency response to severe coastal flooding.

(j) Pollution Governance

(i) Mapping Ship-Based Oil Spills to Estimate the Hazard That Maritime Transportation Represents to Atlantic Basin Coasts

The AtlantOS "oil spill hazard mapping and disaster risk reduction best practices" use case developed a large ensemble oil spill simulation experiment to guide oil spill risk assessments and emergency management. Risk maps can now be rapidly produced, within five minutes, to help evaluate an oil spill hazard (S-Neves et al, 2018). The hazard/risk mapping and production of oil spill hazard bulletins in the Atlantic are generated in an open-access oil spill hazard map portal called Glamor. In emergencies, the users can map the oil spill hazard spatial distribution for any Atlantic coastal area based on the average oil spill concentration found at the coastline. The product provides a decision support tool to coastal managers and governments to help guide emergency clean-up operations and decisions on the allocation of resources to targeted high-risk areas.

(ii) Overall Planning of Land and Sea to Strictly Control Pollutants

China appoints local government heads as river chiefs and bay chiefs across the nation to clean up and protect its water resources. Shandong proposed a new pollution prevention idea of "governing river first, treating land and sea in a coordinated way." In Shandong, 15 rivers flowed into Laizhou Bay. To implement the responsibility of management and protection, Laizhou established a three-level "river chief" organization system of city, town, and village. A total of 13 city-level "river chiefs," 44 town-level "river chiefs," and 662 village-level "river chiefs" were established. Since 2017, Laizhou has concentrated on the comprehensive management of the Baisha River, Nanyang River, and Zhenzhu River basins. A total of 42 million yuan has been invested, about 25,000 meters of silt has been cleared up, more than 30 sewage outlets have been closed, and 64 polluted aquaculture households have been banned. After the linkage of the "river chief" and the "bay chief," the problem of repeated pollution in the Bay has been fundamentally solved (People's Daily, 2018). The current humanitarian payments to unemployed people in Nigeria could be easily directed to areas like this to engage the people.

Chapter 8

Benefits of the Blue Economy

KEY POINTS

i) The Blue economy accounts for about 70% of oxygen provided by the earth's green economy
ii) Over 90% of world transportation and logistics are carried out in the blue space
iii) The global Blue Economy has an asset base of over $24 trillion
iv) Summarily, the blue economy sector accounts for ninety-five percent of US international trade.

8.1 Background

Much had been said about the economic importance of the blue economy in the earlier chapters of this text. It is pertinent that we dedicate some discussion specifically to address these benefits. The chapter, therefore, puts into perspective the benefits that the blue economy offers to the environment and the economy.

8.2 The Benefits

8.2.1 The Environment

(i) Scientists estimate that **50-80%** of the oxygen production on Earth comes from the ocean. The majority of this production is from oceanic plankton-drifting plants, algae, and some bacteria that can photosynthesize (Hahana, 2021).

Apart from the fact that oxygen is a major source of life support to mammals, it is essential to numerous biological systems and processes, such as being a key element in the proteins, sugars, nutrients, and other chemicals that sustain life.

Oxygen also absorbs the harmful ultraviolet rays of the sun and protects the earth's environment and living organisms, or the biosphere (http://unacademy.com)

(ii) The diverse wildlife species and other life forms that make up about 50% of life on planet Earth habit in water bodies such as oceans, seas, rivers, lakes, marshes, and bays. The water bodies are estimated to host about 2.2 million plant species (oceanpolicy.com).

8.2.2 The Economy

(i) It is interesting to note that more than 90% of trade and logistics transportation in the world uses sea routes. For heavy equipment that cannot be transported by air and where roads are not available, especially in transcontinental connections, water sources become the available transportation and haulage source, providing job opportunities for millions of people.

(ii) Globally, the Blue Economy has an asset base of over $24 trillion and it is said to generate at least $2.5 trillion each year from the combination of fishing and aquaculture, shipping, tourism, and other activities. Coastal tourists alone are estimated at six £6 billion. It is estimated that 80% of global trade by volume is carried by the sea. Also, 350 million jobs worldwide are linked to fisheries. It is estimated that by 2025, 34% of crude oil production will come from offshore fields (thecommonwealth.org, 2021)

(iii) Particularly, the United States of America maritime transport carries 95% of the nation's foreign trade. Ships move 2 billion tons of freight in and out of U.S. ports yearly (Oceans Impact the Economy, oceanpolicy.com)

(iv) The livelihood of the majority of people in the world depends on the blue economy. This economy has generated a tremendous amount of support through employment and other life sustenance mechanisms like the provision of food through its fishery products, relaxation through tourism sites, and natural resources for wealth creation.

Chapter 9

Shipbuilding

KEY POINTS

i) The process of building ships and other floating vessels is known as shipbuilding while the location where the construction or repair of ships is done is called a shipyard

ii) Historically, shipbuilding spans the 4th Millennium BC through the current modern era

iii) The first regular oceangoing vessels were built in Western Europe with the advent of the carrack, or sailing sheep, which dominated the seas from the 14th to the 17th century. These ships increased to an unparalleled level of size, complexity, and cost in a very short period.

iv) Contemporarily, China, South Korea, and Japan accounted for the majority of completed gross tonnage in the global shipbuilding industry. The bulk of the global market is supplied with large container, bulk carrier, tanker, and Rwoo o-ro ship supplies by China State Shipbuilding Corporation

9.1 Background

Shipbuilding is the construction of ships and other floating vessels. Ships are built in a facility known as a shipyard. Artisans that build ships are called shipbuilders or shipwrights.

Ships or floating vessels have been in recorded history from the 6th to the 5th millennium BC. This book considers ships and shipbuilding as a very important sector in the Blue Economy. Without ships, activities in the Blue Economy, especially the transportation of goods and services, fishing, mining activities as well as research would have been nearly impossible. We will thus devote this chapter to discussing the entire spectrum of the evolution of ships and shipbuilding in the world.

9.2 History of Ship Building

The original ship-building history available to us is from the 6th to 5th millennium BC of the Ubaid period of Mesopotamia. Ships then were mostly made from bundled reeds coated in bitumen had bipod masts and were sailing in shallow coastal waters of the Persian Gulf (Carter, 2006).

This chapter is dedicated to discussing the whole facet of shipbuilding in the periods of its historical evolution.